

EXCEL MINI PROJECT

Superstore Sales Analysis

Database · 20 Formula Solutions · Final Report

2,538 US retail order records (2014–2017) analysed across 20 states with Excel formulas — from raw data to business recommendations.

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Source workbook: Mini_Project_completed.xlsx

1 · The Database at a Glance

Sheet 1 holds one clean row per order line — who bought, where it shipped, what was sold, and the profit or loss.

2,538

Order records

20

Data fields

764

Unique customers

48

US states

3

Categories

17

Sub-categories

Key fields: Order ID & dates · Ship Mode · Customer & Segment · State & Region · Category, Sub-Category & Product · Sales, Quantity, Discount, Profit

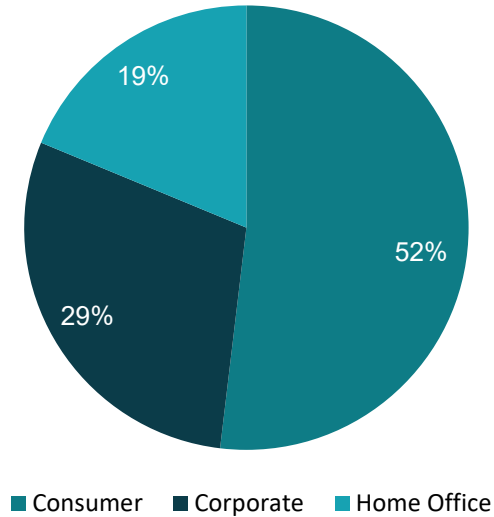
Sample Records from Sheet 1

Order ID	Customer	Segment	State	Region	Category	Sub-Cat	Sales \$	Qty	Disc.	Profit \$
CA-2014-103800	Darren Powers	Consumer	Texas	Central	Office Sup.	Paper	16.45	2	0.20	5.55
CA-2014-106054	Jack O'Briant	Corporate	Georgia	South	Office Sup.	Art	12.78	3	0.00	5.24
CA-2014-109232	Natalie DeCherney	Consumer	S. Carolina	South	Furniture	Chairs	545.94	6	0.00	87.35
CA-2014-149524	Brendan Sweed	Corporate	Pennsylvania	East	Furniture	Bookcases	61.96	4	0.50	-53.29
CA-2014-103366	Erica Hackney	Consumer	Georgia	South	Technology	Accessories	149.95	5	0.00	65.98
US-2014-147774	M. MacIntyre	Corporate	Tennessee	South	Office Sup.	Binders	67.19	1	0.70	-51.52
CA-2014-131387	Arianne Irving	Consumer	California	West	Technology	Phones	1,679.96	5	0.20	126.00

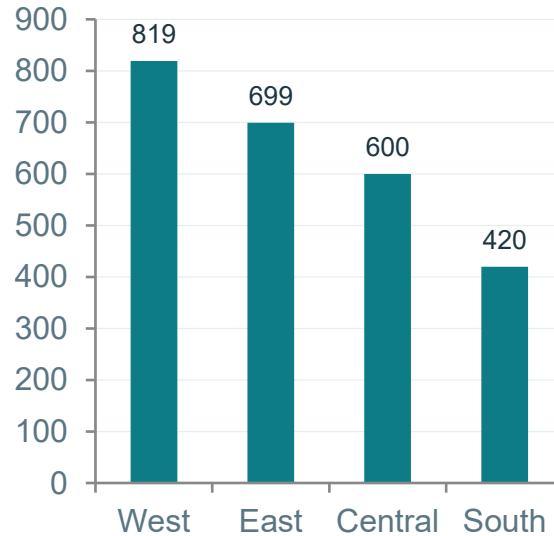
Each row is one order line: who bought, where it shipped, what was sold, and the money made or lost — note how deep discounts (50–70%) push some lines into a loss (red).

How the Data Is Composed

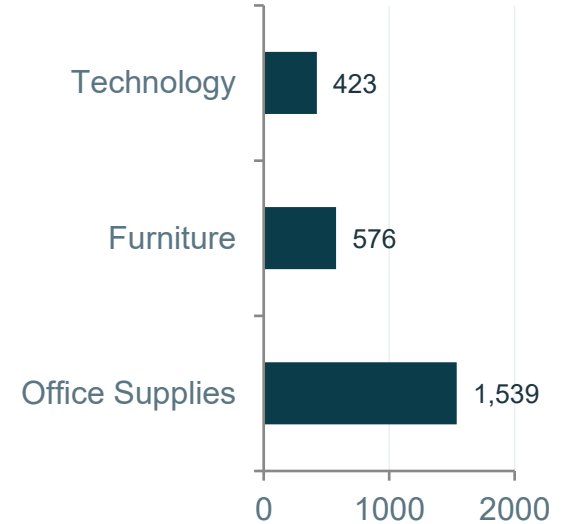
Orders by Segment



Orders by Region



Orders by Category



Consumers place over half of all orders; the West leads by volume; Office Supplies dominates order count while Furniture leads sales value.

2 · Questions & Formulas (Q1 – Q10)

Each formula is written for row 2 (Texas) on the Questions sheet and filled down for all 20 states.

Q#	Question	Formula used
Q1	Region of each State	<code>=INDEX (Sheet1!\$J\$2:\$J\$2539,MATCH (A2,Sheet1!\$I\$2:\$I\$2539,0))</code>
Q2	CONCAT (Region : State)	<code>=CONCATENATE (B2," : ",A2)</code>
Q3	Total Orders per state	<code>=COUNTIF (Sheet1!\$I\$2:\$I\$2539,A2)</code>
Q4	Total Sale per state	<code>=SUMIF (Sheet1!\$I\$2:\$I\$2539,A2,Sheet1!\$P\$2:\$P\$2539)</code>
Q5	Rank by Sale Qty	<code>=RANK (E2,\$E\$2:\$E\$21)</code>
Q6	State Contribution %	<code>=E2/SUM (\$E\$2:\$E\$21)</code>
Q7	Corporate segment sales	<code>=SUMIFS (Sheet1!\$Q\$2:\$Q\$2539,Sheet1!\$I\$2:\$I\$2539,\$A2,Sheet1!\$G\$2:\$G\$2539,H\$1)</code>
Q8	Consumer segment sales	Same SUMIFS with I\$1 (filled across)
Q9	Home Office segment sales	Same SUMIFS with J\$1 (filled across)
Q10	Max Segment by Sale Qty	<code>=INDEX (\$H\$1:\$J\$1,MATCH (MAX (H2:J2),H2:J2,0))</code>

Questions & Formulas (Q11 – Q20)

Q#	Question	Formula used
Q11	1st Order Date	=MINIFS (Sheet1!\$C\$2:\$C\$2539, Sheet1!\$I\$2:\$I\$2539, A2)
Q12	Last Order Date	=MAXIFS (Sheet1!\$C\$2:\$C\$2539, Sheet1!\$I\$2:\$I\$2539, A2)
Q13	Days from Last Order	=DATEDIF (M2, TODAY (), "Y") & " Years " & DATEDIF (M2, TODAY (), "YM") & " Months " & DATEDIF (M2, TODAY (), "MD") & " Days "
Q14	Last Order as "MMM-YY"	=TEXT (M2, "MMM-YY")
Q15	Total Sales per state	=SUMIF (Sheet1!\$I\$2:\$I\$2539, A2, Sheet1!\$P\$2:\$P\$2539)
Q16	Avg Sale per Customer	=P2/SUMPRODUCT ((Sheet1!\$I\$2:\$I\$2539=A2) /COUNTIFS (Sheet1!\$I..I, Sheet1!\$I..I, Sheet1!\$F..F, Sheet1!\$F..F))
Q17	Total Profit per state	=SUMIF (Sheet1!\$I\$2:\$I\$2539, A2, Sheet1!\$S\$2:\$S\$2539)
Q18	Profit After Tax (18%)	=IF (R2<0, R2, IF (R2>0, R2- (18%*R2)))
Q19	Profit Slab <10K / 10–20K / >20K	=IF (R2<10000, "<10K", IF (R2<=20000, "10–20K", ">20K"))
Q20	Highlight values <10K in red	=R2 + Conditional Formatting ► Highlight Cells ► Less Than 10000

Toolkit: INDEX·MATCH · COUNTIF · SUMIF/SUMIFS · MINIFS/MAXIFS · RANK · date functions · nested IF · conditional formatting

3 · Final Report — Executive Summary

\$438.1K

Total sales across the 20
analysed states

2,217

Orders placed in those states

\$37.5K

Total profit before tax

\$30.8K

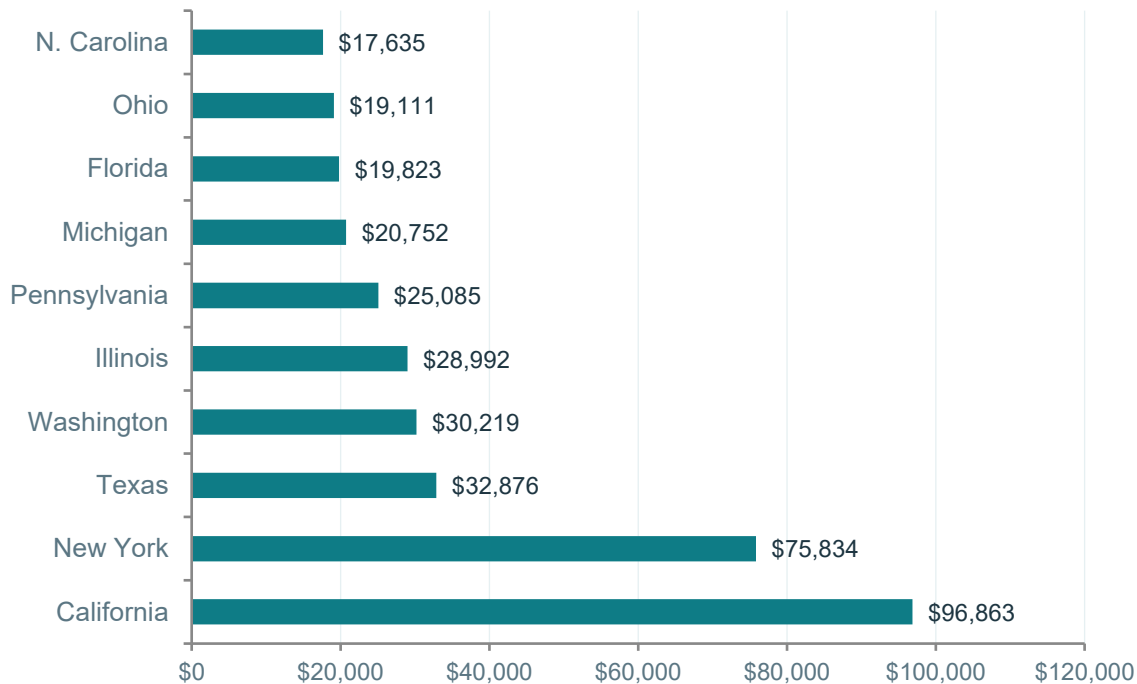
Profit after 18% tax

California and New York carry the business. Together they contribute ~39% of sales and are the only two states in the 10–20K profit slab — every other state sits below 10K, and six states lose money outright.

Consumer is the dominant segment. It generates \$231.4K of the \$438.1K total (52.8%), leading in 16 of the 20 states, versus \$138.2K Corporate and \$68.5K Home Office.

Where the Sales Come From

Top 10 states by total sales



#1 California

\$96.9K — 22.1% of all sales, 522 orders

#2 New York

\$75.8K — 17.3% share, 292 orders

Top-5 share

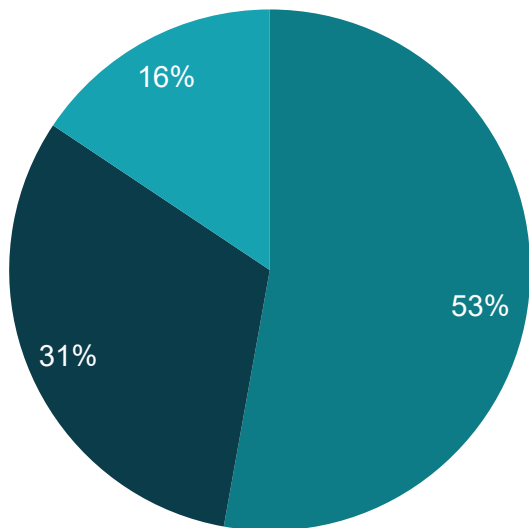
CA + NY + TX + WA + IL $\approx$ 60% of sales

Long tail

Bottom 10 states add just ~\$60K combined

Segment Mix: Consumer Leads Everywhere

Sales by segment (\$438.1K total)



■ Consumer ■ Corporate ■ Home Office

Consumer — \$231.4K (52.8%)

Top segment in 16 of 20 states, including California (\$45.2K) and New York (\$41.8K).

Corporate — \$138.2K (31.5%)

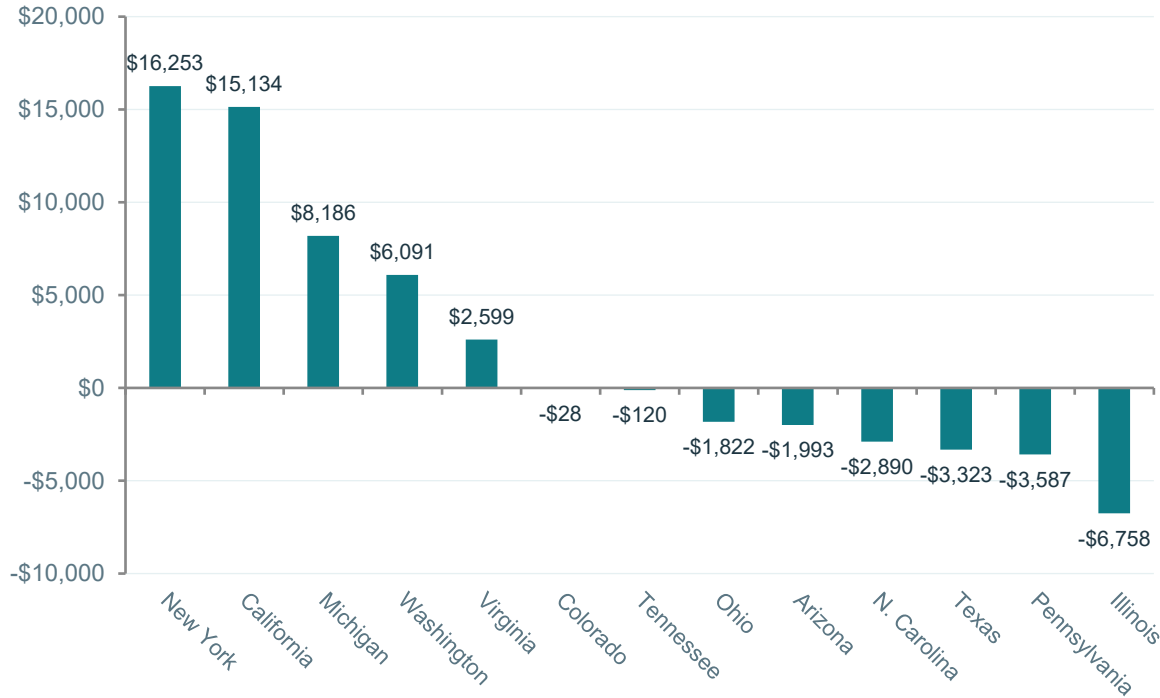
Wins only in Indiana, Massachusetts, New Jersey and Georgia — all smaller markets.

Home Office — \$68.5K (15.6%)

Never the leading segment; strongest in California (\$17.2K) and New York (\$12.1K).

Profitability: Winners and Losers

State profit — best and worst (before tax)



Profit engines

NY (+\$16.3K) and CA (+\$15.1K) deliver 84% of total profit; Michigan quietly earns +\$8.2K on modest sales.

Loss makers

Six states lose money. Illinois (−\$6.8K), Pennsylvania (−\$3.6K) and Texas (−\$3.3K) erase ~\$13.7K — heavy discounting is the main driver.

Profit Slabs & Order Recency

Slab	States	Meaning
10–20K	New York, California	Only two states clear the 10K profit bar
<10K	Remaining 18 states	Flagged red by conditional formatting; incl. 6 loss-making states
>20K	None	No state reaches the top slab

After 18% tax: total profit falls from \$37.5K to \$30.8K. New York keeps \$13.3K and California \$12.4K after tax.

Order dates: first orders arrive in early 2014 (Texas: 03-Jan-2014) and 19 of 20 states place their last order in Nov–Dec 2017 (“Dec-17”) — the customer base stayed engaged for the full four years.

Conclusions & Recommendations

1

Protect the champions

California and New York drive ~39% of sales and 84% of profit — prioritise stock, service and retention there.

2

Fix the discount leak

Illinois, Texas and Pennsylvania sell well but lose money; review the 50–80% discount lines that turn sales into losses.

3

Grow Consumer further

The Consumer segment wins in 16 of 20 states — targeted promotions here have the widest reach.

4

Lift the middle tier

Michigan proves mid-size states can be highly profitable (\$8.2K on \$20.8K sales) — replicate its low-discount mix.

Thank You!

Database · 20 Formula Solutions · Final Report — all built in Excel.

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Questions & feedback welcome